TRoss SEQ

16-STEP MIDI DRUM SEQUENCER · USER MANUAL



Version 0.4.0 · AU / VST3

Ticco Ross · 2026

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1 What TRoss Seq does

TRoss Seq is a 16-step drum sequencer that plays your hardware drum machines over MIDI, locked to your DAW's tempo.

- **13 drum lanes plus an accent row**, 16 steps each, programmed on lit step pads.
- **Every lane sends its own MIDI note on its own channel**, so one TRoss Seq can play one drum box or several at once.
- **12 pattern slots**. A new slot always waits for the current pattern to finish, so changes land on the bar.
- **A drum-aware randomiser** that writes patterns the way each drum is usually played, with styles, density, lane locks, variations, fills and undo.
- **Two ways out**: straight to a MIDI port on your computer, or to your DAW's MIDI routing.
- **Flams** on any step with **Shift**-click, and a **drum map library** to keep one map per drum box.
- Swing, gate, accent velocity, MIDI clock out, MIDI learn, keyboard recording and MIDI file export.

TRoss Seq makes no sound of its own. It is an instrument plug-in that outputs MIDI, which is why it loads on a MIDI or instrument track.

2 Installing

macOS

1. Open **TRossSeq-0.4.0-macOS.pkg** (the manual is in the same download).
2. The installer isn't signed, so the first time macOS blocks it. Open **System Settings → Privacy & Security**, scroll down and click **Open Anyway**, then open the .pkg again.
3. Follow the installer. It puts the Audio Unit in `/Library/Audio/Plug-Ins/Components` and the VST3 in `/Library/Audio/Plug-Ins/VST3`. Click **Customize** to install only one of them.
4. In Ableton Live, open **Settings → Plug-Ins**, check that Audio Units (and/or VST3) are switched on, then hold **Option** and click **Rescan**.

You'll find it in Live's browser under **Plug-Ins → Audio Units → Ticco Ross → TRoss Seq** (or under VST3).

Windows

Run **TRossSeq-0.4.0-Windows-Setup.exe**. If SmartScreen appears, click **More info → Run anyway**. The VST3 goes to `C:\Program Files\Common Files\VST3`. Rescan plug-ins in your DAW.

Linux

Unpack the `.tar.gz` and run `./install.sh`. It copies the VST3 to `~/.vst3` and the LV2 to `~/.lv2`, with no root needed. `./uninstall.sh` removes them.

Updating

Install the new version over the old one, then rescan. Your projects keep their patterns and settings. The version number is shown at the top right of the panel.

3 Quick start in Ableton Live

1. Connect your drum machine to your computer: a MIDI interface (DIN cable to the drum box's MIDI IN), or USB if the drum box has USB MIDI.
2. Create a **MIDI track** and drop **TRoss Seq** on it.
3. Under **MIDI OUT PORT**, pick your interface or drum box. The line underneath should read **CONNECTED**.
4. Open **PRESETS** and pick the map that matches your drum box, or set each lane's **NOTE** and **CH** by hand (section 6). Most drum boxes listen on channel 10.
5. Press play in Live. The step lights run along with Live's transport.
6. Click pads to program a beat, or press **RANDOM**.

> **Why the MIDI OUT PORT?** Live doesn't pass MIDI that comes out of AU or VST3 plug-ins on to other tracks or to MIDI ports. TRoss Seq therefore sends straight to the port itself. Leave **HOST OUT** on or off as you like in Live; it has no effect there.

4 The panel at a glance



TRoss Seq 0.4.0. The window can be resized from its bottom-right corner.

A Feel

VELOCITY, ACCENT, GATE, SWING and FLAM knobs (section 11). Turn a knob to see its value; double-click resets it.

B MIDI output

MIDI OUT PORT selector and status line, HOST OUT, CLOCK OUT, MIDI THRU and PORT DELAY (section 5).

C Drum map

PRESETS: built-in maps and your own saved maps. The line below shows the current map, with *** EDITED** when it has changed (section 6).

D Display

Shows the playing pattern and what's queued next. In a DAW TRoss Seq follows the host's transport and tempo.

E Lanes

Name strips (play, rename, lock), step pads, and per lane NOTE, CH, M (mute / solo) and LEARN.

F Patterns & transport

12 pattern buttons, LENGTH, RUN, CLEAR, REC and EXPORT MIDI.

G Randomiser

DENSITY, RANDOM STYLE, and the RANDOM, FILL and UNDO keys (section 12).

H HELP

Top right. Switches the tips that pop up when you hover a control on or off. Remembered for every copy of TRoss Seq.

Display messages

DISPLAY	MEANING
`PATTERN 1`	Pattern 1 is playing and is the one you're editing.
`PAT 1 > 2`	Pattern 2 is queued. It takes over when pattern 1 is back at step 1.
`PAT 1 > RND`	A random pattern (or fill / variation) is waiting for step 1.

5 Getting the notes to your drum machine

There are two outputs. Use one or both, but don't send both to the same drum box, or every note plays twice.

OUTPUT	HOW IT WORKS	USE IT IN
MIDI OUT PORT	TRoss Seq sends straight to the port you pick: a MIDI interface, or a drum box with USB MIDI. It has its own high-priority timing, so notes don't jitter with the DAW's buffer.	Every DAW. The way to go in Ableton Live.
HOST OUT	The notes go to the DAW's MIDI output for the plug-in, and you route them from there.	DAWs that route plug-in MIDI: Bitwig, Reaper, Logic, Cubase, Studio One…

MIDI OUT PORT

Click the selector to see the ports connected right now; the list refreshes each time it opens. **NONE** switches the port output off. The status line below tells you what's going on:

STATUS	MEANING
CONNECTED	Notes are going to the port.
NOT CONNECTED…	The project asks for a port that isn't plugged in. Plug it in and pick it again, or pick another port.
COULDN'T OPEN…	Another program holds the port (common on Windows; see section 16).

PORT DELAY

Delays everything sent to the port by 0–250 ms, to line the drum box up with the audio coming out of your DAW. Start at about your audio buffer size (in ms) and adjust by ear against a sound in the DAW.

CLOCK OUT

Sends MIDI clock (24 ticks per quarter note), Start, Stop, Continue and Song Position to the port. Switch it on if your drum box has tempo-synced effects or its own sequencer that should follow along, and set the drum box to sync to external / MIDI clock. Clock goes to the port only.

MIDI THRU

Passes the MIDI coming into the track (your keyboard or pads) through to the outputs, so you can play the drum box live through TRoss Seq. In Live, arm the track (or set its monitor to **In**) so your keyboard reaches the plug-in.

6 Tracks: names, notes, channels

Each of the 13 lanes has a name strip on the left and its MIDI settings on the right.

CONTROL	WHAT IT DOES
Name strip	**Click** to play the lane's note now. **Shift**-click plays it at accent velocity. It lights up every time the lane sends a note.
Rename	**Double-click** the name and type up to 10 characters; **Return** keeps it, **Esc** cancels. The name also tells the randomiser what kind of drum the lane is (section 12).
NOTE	The MIDI note the lane sends. Drag the red read-out up or down. Note names follow Ableton's convention (C1 = 36).
CH	The MIDI channel, 1–16. Drag up or down.
LEARN	Click it (it blinks), then hit a key or pad: the lane takes that note and channel.
Right-click a name	Rename, Learn, Randomise this lane, Lock from the randomiser.

Drum maps and the PRESETS menu

A **drum map** is the set of lane names, notes, channels and locks. Make one for each drum box you own, save it once, and pick it whenever you use that box. The line under PRESETS shows the current map, e.g. **MAP: TR-8S STUDIO**, with an orange *** EDITED** once you change anything after picking or saving it.

PRESETS MENU	WHAT IT DOES
Built-in maps	General MIDI drums, Roland TR-8 / TR-8S, Korg volca beats. Sets all 13 lanes at once (full table in section 15).
My maps	Your saved maps. Click one to load it. A tick shows the current, unedited map.
Save "*name*"	Overwrites the current saved map with your changes (only when it's one of your maps and it's edited).
Save map as...	Saves the current lanes under a new name. If the name exists, you're asked before it's replaced.
Rename map / Delete map	Pick a map from the list. Deleted maps go to the bin / trash.
Show maps folder	Opens `Documents/TRoss Seq/Maps`. Each map is one `.tsmap` file: copy them to back up, or to use on another computer.
All lanes on channel...	Moves every lane to one channel.
Save map + MIDI port as default	New copies of TRoss Seq start with the current map and port.

> **Check your drum box.** The preset maps follow each maker's published default notes. If a sound doesn't trigger, look up its note in the drum box's manual or MIDI settings and fix that lane with LEARN or the NOTE read-out.

Using several drum boxes

Give each drum box its own MIDI channel and set the lanes' CH to match: for example lanes 1–8 on channel 10 for one box, lanes 9–13 on channel 11 for another. Both boxes can sit on the same MIDI port (daisy-chained

through MIDI THRU) or you can use two TRoss Seq instances with a port each.

7 Mute and momentary solo

ACTION	RESULT
Click M	Mutes the lane (the button lights yellow). Click again to unmute.
Shift + hold M	**Momentary solo.** The button turns blue and shows **S**; only that lane plays, even if it is muted itself, and every other lane goes quiet. Let go of the mouse button and everything is back exactly as it was. Nothing is toggled.

Mutes can be automated in your DAW (*Track 1 Mute* … *Track 13 Mute*). They don't affect clicks on the name strips or MIDI THRU.

8 Programming patterns

- **Click a pad** to switch a step on or off. **Drag along a lane** to paint several steps on (or off, if you started on a lit pad).
- The pads are coloured in groups of four, so the beats are easy to see. The LEDs above follow the playing step, and the playing column gets a white outline.
- **ACCENT row:** on any step where it's lit, every lane plays at the ACCENT velocity instead of the VELOCITY.

Flams

A flam is a quick, softer grace note just before the main hit, like a drummer's two sticks landing almost together.

- **Shift**-click a pad to add a flam. If the step was off, it switches on with the flam. The pad shows a small extra LED before the main one.
- **Shift**-click it again to remove the flam; the step stays on. A plain click switches the step off, flam included.
- The grace note plays **before** the step, at 60 % of the step's velocity, so the main hit stays exactly on the grid. The **FLAM** knob sets how far before (section 11).
- Flams work on every lane except ACCENT, are saved with the project and are included in EXPORT MIDI. Flams are only ever set by you: the randomiser never adds them (section 12).

CONTROL	WHAT IT DOES
LENGTH	Pattern length, 1–16 steps. Drag the red read-out up or down; double-click for 16. Steps past the length are dimmed and don't play. Use odd lengths for patterns that drift against the bar.
RUN	Sequencer on/off. With RUN off, the steps still advance but send nothing; MIDI THRU and the name strips keep working.
CLEAR	Clears the pattern that's playing, accent row included. (RANDOM's UNDO doesn't bring a cleared pattern back.)

TRoss Seq always follows the DAW: it starts, stops, loops and jumps with the transport, and step 1 falls on the downbeat of each bar. When the transport stops, any note still held gets its note-off.

9 Recording from a keyboard or pads

With REC on (it is by default) and the song playing, every incoming note that matches a lane is written into the nearest step of the playing pattern. Swing is taken into account.

- A note matches a lane by **note and channel** first, then by **note alone**, so a keyboard on channel 1 still records into lanes that send on channel 10.
- Velocity above 100 also sets the accent on that step.
- Clicking a name strip while playing records too (**Shift**-click for an accent).
- While the transport is stopped, nothing is recorded; hits just play.

Switch REC off to play along without writing anything.

10 The 12 pattern slots

- **Click a pattern button** to select it. While the song plays, the new slot waits until the current pattern reaches step 1 (its LED blinks while queued), so changes always land cleanly. While stopped, it switches at once.
- The LED shows the slot that's playing; the light cap shows the selected one. A small dash marks empty slots.
- **Right-click a pattern button** to **Copy to** another slot, or **Clear** it.
- The *Pattern* parameter can be automated, so you can arrange pattern changes in your DAW's timeline.

11 Feel: velocity, accent, gate, swing, flam

KNOB	RANGE	WHAT IT DOES
VELOCITY	1–127	Velocity of normal steps. Default 100.
ACCENT	1–127	Velocity of steps with the ACCENT row lit, and of **Shift**-clicked name strips. Default 127.
GATE	5–95 %	How long each note is held, as a fraction of a 16th note. Most drum boxes only care about the note-on; longer gates matter for sounds that respond to note length.
SWING	50–75 %	50 % is straight. Higher values push every second 16th (steps 2, 4, 6…) later: 54–58 % is a light shuffle, 66.7 % a triplet feel, 75 % the maximum.
FLAM	5–80 ms	How far before a flammed step its grace note plays. Default 25 ms; 10–20 ms is tight, 40 ms and up sounds like two separate hits. Never more than half a step.

12 The randomiser

The randomiser writes patterns lane by lane, the way each drum is usually played, instead of throwing random dots on the grid.

CONTROL	WHAT IT DOES
RANDOM	Writes a **whole new pattern** into the playing slot. Every unlocked lane is rewritten over the current LENGTH. **The ACCENT row is never changed.**
Shift-click RANDOM	A **variation**: only a few steps of a few lanes change, and the anchors (kick on step 1, the backbeat) stay. Good for keeping a groove alive without losing it.
FILL	Rewrites only the **last four steps** of the pattern with a fill: toms, snare, hats and percussion. Press it before the bar you want the fill in.
UNDO	Goes back to the pattern as it was before the last RANDOM, variation, fill or lane randomise; up to 32 steps back. Greyed out when there's nothing to undo.
DENSITY	How busy random patterns are. 50 % follows the style as designed; lower is sparser, higher is busier. The anchors (kick on 1, the backbeat in STRAIGHT) are kept at any density.
RANDOM STYLE	The character of what RANDOM and the variation write (below).
Alt-click a name (**Option**-click on a Mac)	**Locks** the lane: a padlock appears and RANDOM, the variation and FILL leave it alone. Alt-click again to unlock. Locks are saved with the project.
Right-click a name → Randomise this lane	Rolls just that one lane, even if it's locked.

Timing

While the song plays, a random result **waits for step 1**, so the pattern never changes in the middle of a bar. The RANDOM key's light blinks and the display shows `PAT 1 > RND` until it takes over. Press RANDOM again while it's waiting to roll a different one; press UNDO to cancel it. While stopped, results are written at once.

Styles

STYLE	CHARACTER
STRAIGHT	Four on the floor: kick on the beats with the odd extra, snare always on the backbeat (steps 5 and 13), 8th-note hats with some 16ths, open hats on the off-beats, toms near the end of the bar.
ELECTRO	Syncopated kick (always on step 1, then pushed around steps 4, 7, 11 and 12), snare on step 13 and usually 5, clap mostly on 13, busy 16th hats, occasional open hats and tom fills.
BROKEN	Scattered and off the grid: kick on 1, then anywhere; snare not tied to the backbeat; looser hats and more percussion.
HOUSE	Kick on every beat, open hat on every off-beat, clap on 5 and 13, shuffled 16th closed hats. Pairs well with SWING.

TECHNO	Driving four-on-the-floor kick, sparse claps, rolling 16th hats, rim and percussion building tension. Darker and busier than HOUSE.
DETROIT	Soulful techno: syncopated hats and claps, cowbell and clave figures, the odd extra kick.
MIAMI BASS	Booming, bouncy kick figures with double hits, snare on 13, fast hats and tom runs.
BREAKBEAT	Funky break: kick on 1 and around 11, snare on 5 and 13 with ghost notes, busy hats with gaps.
HIP-HOP	Laid back: sparse kick, heavy snare on 5 and 13, simple 8th hats. Use it with SWING.
DUB	Very sparse and spacious, with rim shots instead of a snare backbeat and long gaps.
LATIN	Built around the son clave on the claves, rim and cowbell lanes, with toms and agogo interlocking.
INDUSTRIAL	Hard and mechanical: kick on every beat and snare on the backbeat, locked to the grid, few hats.

Flams and the randomiser

The randomiser never adds flams. `RANDOM` and **Randomise this lane** remove the flams on the lanes they rewrite; locked lanes keep theirs, and variations and FILL keep flams on steps that stay on. `UNDO` brings removed flams back.

How lanes are recognised

The randomiser decides what each lane is from its **name** first, so renaming a lane changes how it's treated. If the name doesn't say, it goes by the lane's **General MIDI note**.

TREATED AS	NAMES THAT COUNT	GM NOTES	PLAYS LIKE
Kick	BD, KICK, BASS…	35, 36	Anchored on step 1, plus beats or syncopation
Snare	SD, SNARE	38, 40	Backbeat, ghost notes
Clap	CP, HC, CLAP	39	Backbeat, sometimes
Closed hat	CH, HH, CLOSED, CL HAT, HAT	42, 44	Busy 8ths and 16ths
Open hat	OH, OPEN, OP HAT	46	Off-beats, never on the same step as a closed hat
Cymbal	CY, CC, RD, RC, CRASH, RIDE	49, 51, 52, 53, 55, 57, 59	Rare, mostly on step 1
Tom	TOM, LT, MT, HT	41, 43, 45, 47, 48, 50	Sparse, near the end of the bar
Percussion	RS, RIM, CB, COWBELL, CL, CLAVES, AGOGO, CONGA…	everything else	Sparse off-beat accents

> **Workflow tip.** Program or keep a kick and snare you like, Alt-click to lock them, then press RANDOM until the hats and percussion groove. Shift-click RANDOM for variations, FILL before a change, and UNDO whenever you've gone one roll too far.

13 Exporting MIDI

Click EXPORT MIDI for a menu:

- **Pattern N (playing)** saves the playing pattern as a `.mid` file.
- **All patterns, in a row** saves every non-empty slot one after another in a single file.
- **All patterns as separate files** saves one file per non-empty slot into a folder you choose.

Drag the EXPORT MIDI key onto a MIDI track in your DAW to drop the playing pattern in as a clip.

Exported files use each lane's note and channel, the current VELOCITY, ACCENT, GATE, SWING, FLAM, LENGTH and the song tempo, with 16th-note steps. Flams are written as a grace note before the step.

14 Saving and defaults

Everything is saved with your DAW project: all 12 patterns with their flams, the playing slot, the drum map (and whether it was edited), every knob and switch, and the MIDI port. If the saved port isn't connected when the project opens, the status line says so, and TRoss Seq reconnects to a port with the same name when it's available.

PRESETS → Save map + MIDI port as default stores your lane names, notes, channels, locks and port, and every new copy of TRoss Seq starts with them. Saved drum maps live in `Documents/TRoss Seq/Maps` and are shared by all your projects. The HELP setting is remembered for every copy of TRoss Seq.

15 Reference

Preset note maps

LANE	GENERAL MIDI (CH 10)		ROLAND TR-8 / TR-8S (CH 10)		KORG VOLCA BEATS (CH 10)	
1	BD	36	BD	36	KICK	36
2	SD	38	SD	38	SNARE	38
3	LT	45	LT	43	LO TOM	43
4	HT	50	HT	50	HI TOM	50
5	CP	39	HC	39	CLAP	39
6	CH	42	CH	42	CL HAT	42
7	OH	46	OH	46	OP HAT	46
8	CY	49	CC	49	CRASH	49
9	RS	37	RS	37	CLAVES	75
10	MT	47	MT	47	AGOGO	67
11	RD	51	RC	51	RIM	37
12	CB	56	CB	56	MID TOM	47
13	CL	75	CL	75	RIDE	51

Lanes in grey are spare: those notes are not part of the drum box's own map. Point them at another drum box or reuse them.

Mouse and keyboard

WHERE	ACTION
Step pad	Click: on/off · Drag: paint
	Shift-click: flam on / off
Name strip	Click: play · Shift: accent
	Double-click: rename
	Alt / Option-click: lock
	Right-click: menu
M	Click: mute · Shift + hold: solo

WHERE	ACTION
RANDOM	Click: new · Shift: variation
Pattern button	Click: select · Right-click: copy / clear
Knob	Drag · Double-click: default
Red read-outs	Drag up / down
EXPORT MIDI	Click: menu · Drag: clip
HELP	Hover tips on / off

Automatable parameters

Run · Length · Pattern · Record · Velocity · Accent Velocity · Gate · Swing · Flam Time · Port Delay · MIDI Thru · Host Out · Clock Out · Random Density · Random Style · Track 1–13 Mute

16 Troubleshooting

PROBLEM	WHAT TO CHECK
The drum box plays nothing	Is the song playing and RUN on? Does the MIDI OUT PORT status read CONNECTED? Do the lanes' CH match the channel the drum box listens on (often 10)? Are the notes right for your drum box (PRESETS, LEARN)? Is the lane muted? In Live, remember HOST OUT does nothing; use the port.
Every note plays twice	Both outputs reach the same drum box. Turn HOST OUT off, or set MIDI OUT PORT to NONE.
One sound doesn't trigger	That lane's note isn't what the drum box expects. Click LEARN on the lane and hit the pad on the drum box (if it sends MIDI), or look up the note in its manual.
Port shows COULDN'T OPEN (Windows)	Most Windows MIDI drivers let only one program use a port. Switch that port off in your DAW's MIDI settings, close other MIDI apps, then pick it again in TRoss Seq.
Port shows NOT CONNECTED	The port saved in the project isn't plugged in. Connect it and pick it again, or choose another.
Drum box is early or late against the DAW	Adjust PORT DELAY until they line up.
Drum box's tempo / effects don't follow	Switch CLOCK OUT on and set the drum box to sync to external MIDI clock.
RANDOM doesn't change anything at once	While playing, the result waits for step 1; the display shows `PAT 1 > RND`. Check the lane isn't locked (padlock).
Keyboard doesn't play or record	In Live, arm the track or set monitor to In. MIDI THRU must be on to pass it to the drum box; REC must be on and the song playing to record.
The pop-up tips get in the way	Switch **HELP** off (top right). Knob values still show while you turn them.
A flam doesn't sound like a flam	Some drum boxes cut a sound short when it retriggers quickly, or ignore low velocities. Try a longer FLAM time, or raise VELOCITY.
TRoss Seq doesn't appear in Live	Rescan plug-ins (Option-click Rescan on a Mac). Check Audio Units / VST3 are switched on in Settings → Plug-Ins. On a Mac, check the installer actually ran (Privacy & Security → Open Anyway).